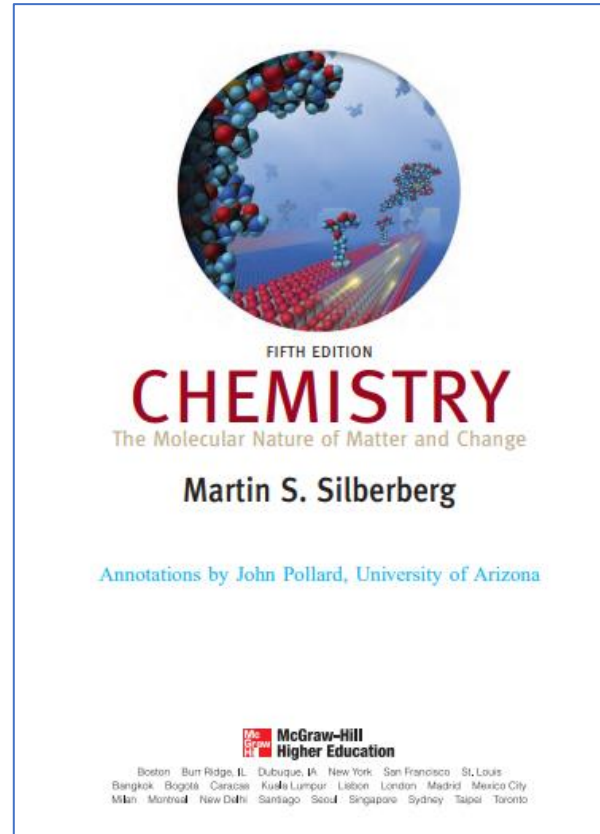


# GENERAL CHEMISTRY – CHE 1201

Chemistry of Elements

*Alkaline Earth Metals*

# Referencing Text



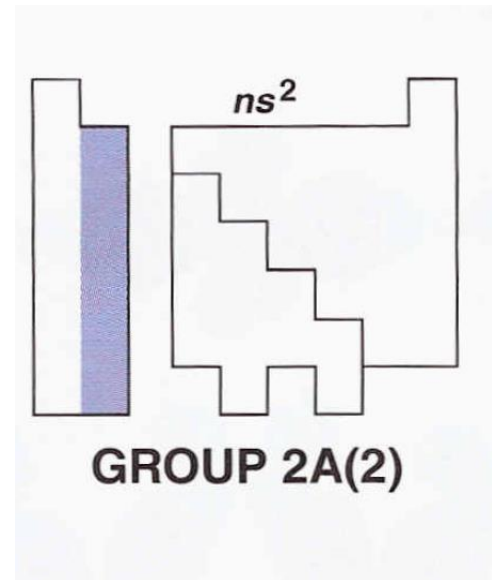
## 14.4 GROUP 2A(2): THE ALKALINE EARTH METALS

The Group 2A(2) elements are called *alkaline earth metals* because

- Their oxides give basic (alkaline) solutions
- Melt at high temperatures that they remained as solids (“earths”).

### How Do the Physical Properties of the Alkaline Earth and Alkali Metals Compare?

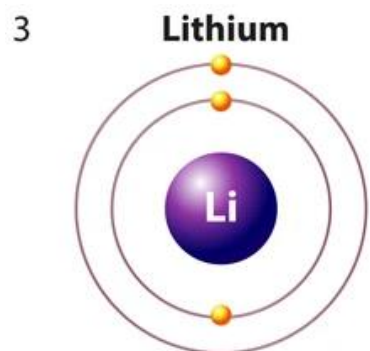
In general terms, the elements in Groups 1A(1) and 2A(2) behave as close cousins. The differences occur between the groups are due to the change in outer electron configuration:  $ns^2$  vs.  $ns^1$ .



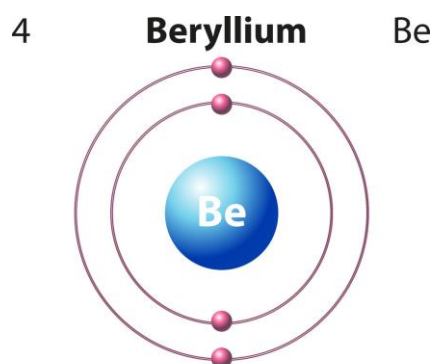
- Two electrons are available from each 2A atom for metallic bonding, and the nucleus contains one additional positive charge.
- These factors make the attraction between delocalized electrons and atom cores greater.
- Consequently, melting and boiling points are much higher for 2A metals than for the corresponding 1A metals
- In fact, the 2A elements melt at around the same temperatures as the 1A elements boil!
- **Compared with transition metals, such as iron and chromium, the alkaline earths are soft and lightweight, but they are much harder and more denser than the alkali metals.**

## How Do the Chemical Properties of the Alkaline Earth and Alkali Metals Compare?

- The alkaline earth metals display a wider range of chemical behavior than the alkali metals, largely because of the behavior of beryllium.



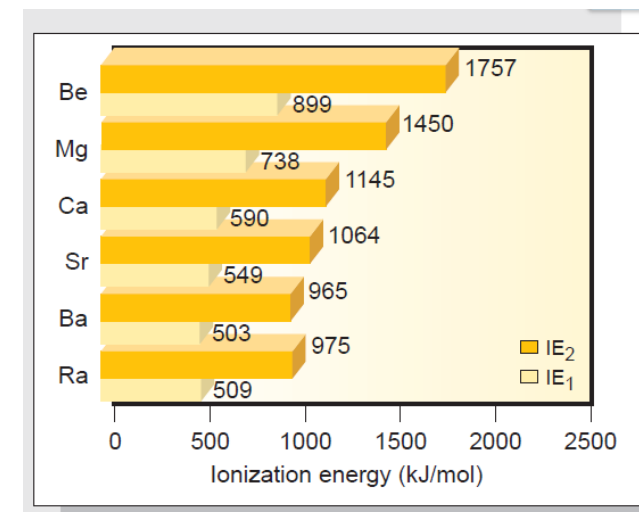
Atomic mass: 6.94  
Electron configuration: 2, 1



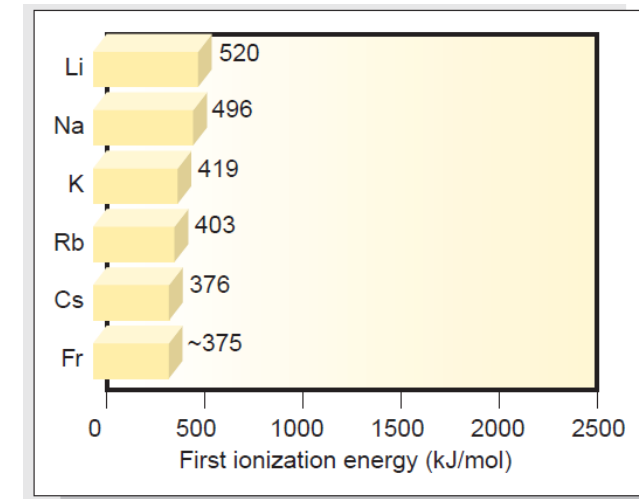
Atomic mass: 9.0121  
Electron configuration: 2, 2

| Atomic radius (pm) |                                                                                     | Ionic radius (pm)    |
|--------------------|-------------------------------------------------------------------------------------|----------------------|
| Be 112             |  |                      |
| Mg 160             |  | Mg <sup>2+</sup> 72  |
| Ca 197             |  | Ca <sup>2+</sup> 100 |
| Sr 215             |  | Sr <sup>2+</sup> 118 |
| Ba 222             |  | Ba <sup>2+</sup> 135 |
| Ra (~220)          |  | Ra <sup>2+</sup> 148 |

| Atomic radius (pm) |                                                                                     | Ionic radius (pm)   |
|--------------------|-------------------------------------------------------------------------------------|---------------------|
| Li 152             |  | Li <sup>+</sup> 76  |
| Na 186             |  | Na <sup>+</sup> 102 |
| K 227              |  | K <sup>+</sup> 138  |
| Rb 248             |  | Rb <sup>+</sup> 152 |
| Cs 265             |  | Cs <sup>+</sup> 167 |
| Fr (~270)          |  | Fr <sup>+</sup> 180 |

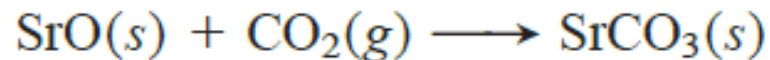


- Because the second valence electron lies in the same sublevel as the first, it is not shielded from the additional nuclear charge very well, and so  $Z_{\text{eff}}$  is greater. Therefore, Group 2A(2) elements have smaller atomic radii and higher ionization energies than Group 1A(1) elements.



## How Do the Chemical Properties of the Alkaline Earth and Alkali Metals Compare?

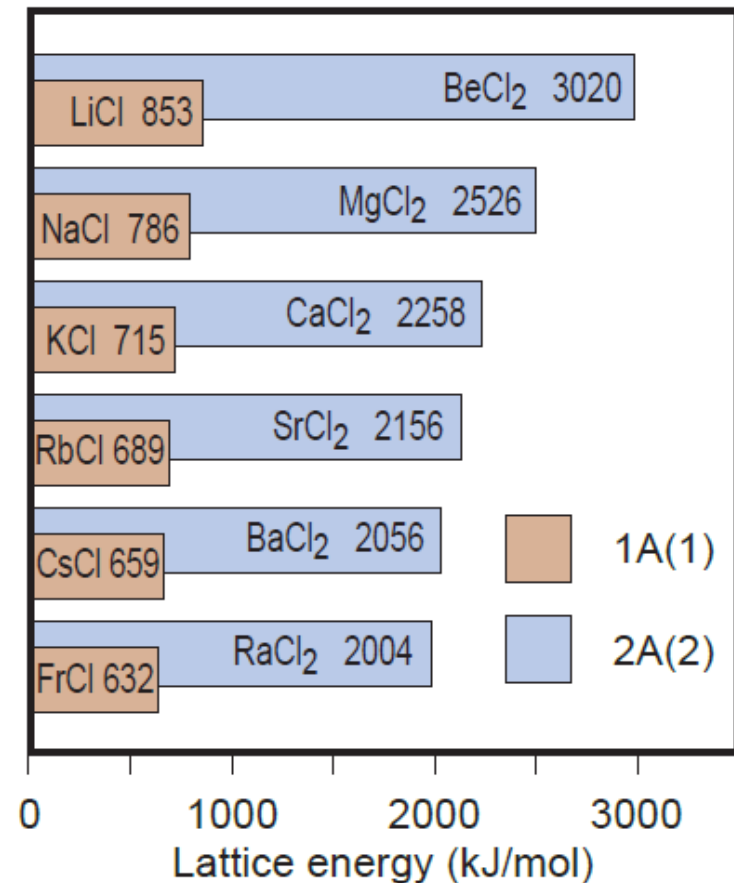
- Despite the higher IEs, *all the alkaline earths (except Be) form ionic compounds as 2 cations.*
- Beryllium behaves differently because so much energy is needed to remove two electrons from this tiny atom that it never forms discrete  $\text{Be}^{2+}$  ions, and its bonds are polar covalent.
- Like the alkali metals, the alkaline earth metals are strong reducing agents. Each element reduces  $\text{O}_2$  in air to form the oxide (Ba also forms the peroxide,  $\text{BaO}_2$ ).
- Except for Be and Mg, which form adherent oxide coatings, the alkaline earths reduce  $\text{H}_2\text{O}$  at room temperature to form  $\text{H}_2$ .
- And, except for Be, they reduce the halogens,  $\text{N}_2$ , and  $\text{H}_2$  to form ionic compounds.
- The Group 2A oxides are strongly basic (except for amphoteric  $\text{BeO}$ ) and react with acidic oxides to form salts, such as sulfites and carbonates; for example,



- Natural carbonates, such as limestone and marble, are major structural materials and the commercial sources for most 2A compounds.

| Atomic radius (pm) |                                                                                     | Ionic radius (pm)       |
|--------------------|-------------------------------------------------------------------------------------|-------------------------|
| Be<br>112          |  |                         |
| Mg<br>160          |  | $\text{Mg}^{2+}$<br>72  |
| Ca<br>197          |  | $\text{Ca}^{2+}$<br>100 |
| Sr<br>215          |  | $\text{Sr}^{2+}$<br>118 |
| Ba<br>222          |  | $\text{Ba}^{2+}$<br>135 |
| Ra<br>(~220)       |  | $\text{Ra}^{2+}$<br>148 |

- Group 2A salts have much higher lattice energies than Group 1A salts because the 2A cations are smaller and doubly charged.
- One of the main differences between the two groups is the lower solubility of 2A salts in water. Their ions are smaller and more highly charged than 1A ions, resulting in much higher charge densities. Therefore, stronger the electrostatic attraction among each other, harder the water molecule to pull them apart.



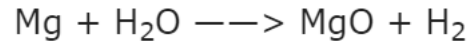
## The Anomalous Behavior of Beryllium (Differ from the rest of the members of the family)

Main reasons are;

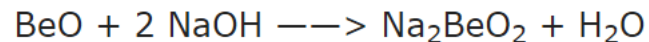
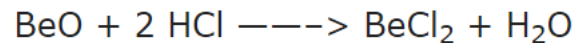
- Exceptionally small atomic and ionic size
- High ionization enthalpy
- Absence of d orbitals in its valence shell

Some important properties in which beryllium differ from rest of the members

- Harder than other members
- Higher boiling and melting points
- Does not react with water even at high temperature, while others do



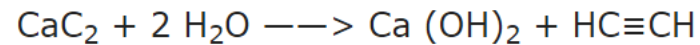
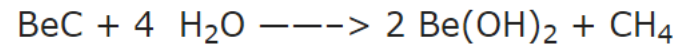
- Forms covalent compounds whereas other members form ionic compounds
- Beryllium oxides and hydroxides are amphoteric while other oxides and hydroxides of other alkaline earth metals are basic.



## The Anomalous Behavior of Beryllium (Differ from the rest of the members of the family)

Some important properties in which beryllium differ from rest of the members

- carbides of Be are covalent whereas carbides of other members are ionic
- Be carbide react with water to produce methane gas, carbides of other members form acetylene gas



- Beryllium does not exhibit coordination number of more than four since its valence shell has only 4 orbitals, one s and p, other elements in the group can show coordination number 6 using s, p and d orbitals



## Diagonal Relationships: Lithium and Magnesium

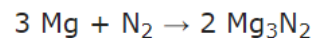
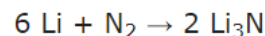
|   | 1A<br>(1) | 2A<br>(2) | 3A<br>(13) | 4A<br>(14) |
|---|-----------|-----------|------------|------------|
| 2 | Li        | Be        | B          |            |
| 3 |           | Mg        | Al         | Si         |

(1) Both have almost **similar electronegativities**.

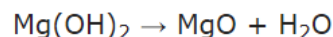
(2) Both Li and Mg are **quite hard**. They are harder and lighter than other elements in their respective groups.

(3) Both LiOH and  $\text{Mg}(\text{OH})_2$  are **weak bases**.

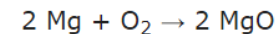
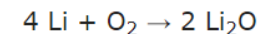
(4) Both **form ionic nitrides** when heated in atmosphere of Nitrogen.



(5) The **hydroxides** of both lithium and magnesium **decompose upon heating**.

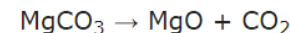
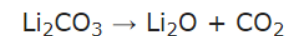


(6) Both lithium and magnesium **combine with oxygen** to form **monoxide** while other members of their respective groups form peroxide and superoxide.



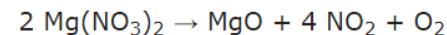
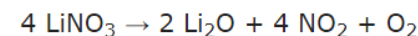
The  $\text{Li}_2\text{O}$  and  $\text{MgO}$  thus formed do not combine with excess  $\text{O}_2$  to form peroxide and superoxide.

(7) The **carbonates** of these metals **decompose on heating** to the corresponding **oxides** with the evolution of **carbon dioxide**.



(8) Both Lithium and magnesium **do not form solid bicarbonates**.

(9) Both Lithium and magnesium **nitrate decompose on heating** producing nitrogen dioxide.



(10) The hydroxides, carbonate, oxalates, phosphates and fluorides of both Lithium and magnesium are sparingly soluble in water.

**KEY**

Atomic No.  
Symbol  
Atomic mass  
Valence e<sup>-</sup>  
configuration  
Common oxidation states

$ns^2$

GROUP 2A(2)

|                                   |                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------|
| 4<br>Be<br>9.012<br>$2s^2$<br>+2  |   |
| 12<br>Mg<br>24.30<br>$3s^2$<br>+2 |   |
| 20<br>Ca<br>40.08<br>$4s^2$<br>+2 |   |
| 38<br>Sr<br>87.62<br>$5s^2$<br>+2 |   |
| 56<br>Ba<br>137.3<br>$6s^2$<br>+2 |  |
| 88<br>Ra<br>(226)<br>$7s^2$<br>+2 | No sample available                                                                |

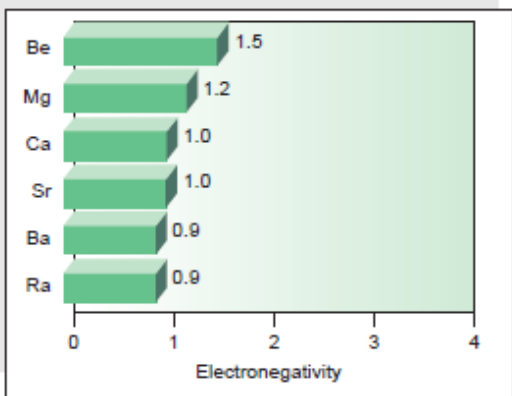
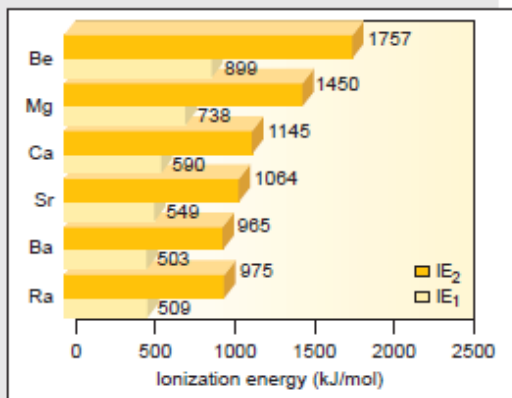
## Atomic Properties

Group electron configuration is  $ns^2$  (filled  $ns$  sublevel). All members have the +2 oxidation state and, except for Be, form compounds with an  $E^{2+}$  ion.

Atomic and ionic sizes increase down the group but are smaller than for the corresponding 1A(1) element.

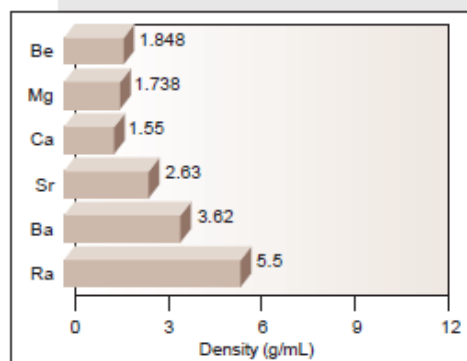
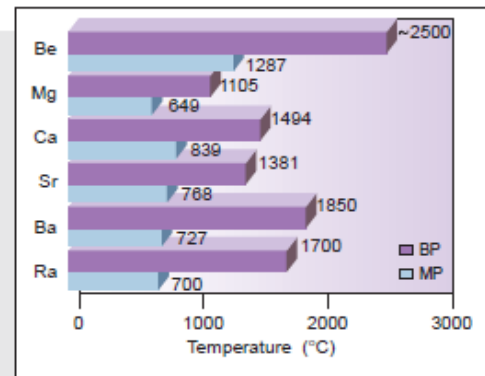
IE and EN decrease down the group but are higher than for the corresponding 1A(1) element.

| Atomic radius (pm) |                                                                                   | Ionic radius (pm) |
|--------------------|-----------------------------------------------------------------------------------|-------------------|
| Be<br>112          |  |                   |
| Mg<br>160          |  | $Mg^{2+}$<br>72   |
| Ca<br>197          |  | $Ca^{2+}$<br>100  |
| Sr<br>215          |  | $Sr^{2+}$<br>118  |
| Ba<br>222          |  | $Ba^{2+}$<br>135  |
| Ra<br>(~220)       |  | $Ra^{2+}$<br>148  |



## Physical Properties

Metallic bonding involves two valence electrons. These metals are still relatively soft but are much harder than the 1A(1) metals.

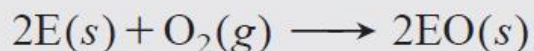


Melting and boiling points generally decrease, and densities generally increase down the group. These values are much higher than for 1A(1) elements, and the trend is not as regular.

## Important Reactions

The elements (E) act as reducing agents in reactions 1 to 5; note the similarity to reactions of Group 1A(1). Reaction 6 shows the general basicity of the 2A(2) oxides; reaction 7 shows the general instability of their carbonates at high temperature.

1. The metals reduce  $O_2$  to form the oxides:

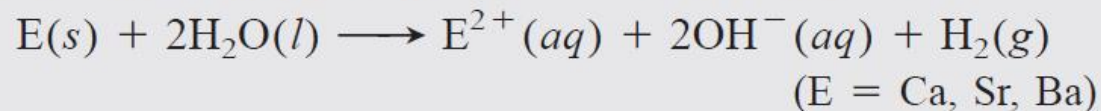


Ba also forms the peroxide,  $BaO_2$ .



Magnesium ribbon burning

2. The larger metals reduce water to form hydrogen gas:

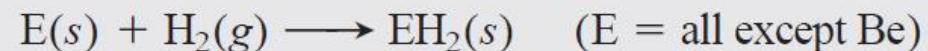


Be and Mg form an adherent oxide coating that allows only slight reaction.

3. The metals reduce halogens to form ionic halides:



4. Most of the elements reduce hydrogen to form ionic hydrides:



5. Most of the elements reduce nitrogen to form ionic nitrides:



6. Except for amphoteric  $BeO$ , the element oxides are basic:



$Ca(OH)_2$  is a component of cement and mortar.

7. All carbonates undergo thermal decomposition to the oxide:



This reaction is used to produce  $CaO$  (lime) in huge amounts from naturally occurring limestone (*margin note, p. 574*).



## Important Compounds



1. Beryl,  $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ . Beryl, the industrial source of Be metal, also occurs as a gemstone with a variety of colors (*see photo*). It is chemically identical to emerald, except for the trace of  $\text{Cr}^{3+}$  that gives emerald its green color.

2. Magnesium oxide,  $\text{MgO}$ . Because of its high melting point ( $2852^\circ\text{C}$ ),  $\text{MgO}$  is used as a refractory material for furnace brick (*see photo*) and wire insulation.

Industrial kiln



3. Alkylmagnesium halides,  $\text{RMgX}$  ( $\text{R}$  = hydrocarbon group;  $\text{X}$  = halogen). These compounds are used to synthesize many organic compounds. Organotin agricultural fungicides are made by treating  $\text{RMgX}$  with  $\text{SnCl}_4$ :



4. Calcium carbonate,  $\text{CaCO}_3$ . Occurs as enormous natural deposits of limestone, marble, chalk (*see photo*), and coral. Used as a building material, to make lime, and, in high purity, as a toothpaste abrasive and an antacid.

White cliffs,  
Dover, England

